Supplementary Figure 1 | Detailed network framework of CELLECT. The overall framework of the CELLECT model is illustrated; The upper-left section depicts the main network structure, a 3D U-Net, with data dimensions and arrows representing the data flow and connection between modules. The color of the arrows corresponds to the network modules shown in the at the left bottom panel. The main model generates 3 outputs: cell division predictions, a confidence map, and a 64-dimensional embedding map, all maintaining same spatial resolution as the input, derived from 2 frames of input images. The confidence map serves as the basis for cell segmentation and size estimation. The lower-right section introduces the Center Enhancement Network (CEN), a light-weight 3D U-Net model. This network takes the confidence map and the 2 original image frames as inputs, refining the probability distribution by focusing on high-probability regions around cell center points while suppressing probabilities in other areas. The peaks in the confidence map generated by the CEN correspond directly to precise center point locations. This framework enables high-accuracy cell division prediction, segmentation, and size estimation while significantly improving the precision of center point localization. The network outputs shown here were generated from the Cell Tracking Challenge (CTC) Fluo-N3DH-CE dataset. The model used was trained on sequence #3 of the mskcc-confocal dataset.